1.3 – 1.4 The Integral and Comparison Tests

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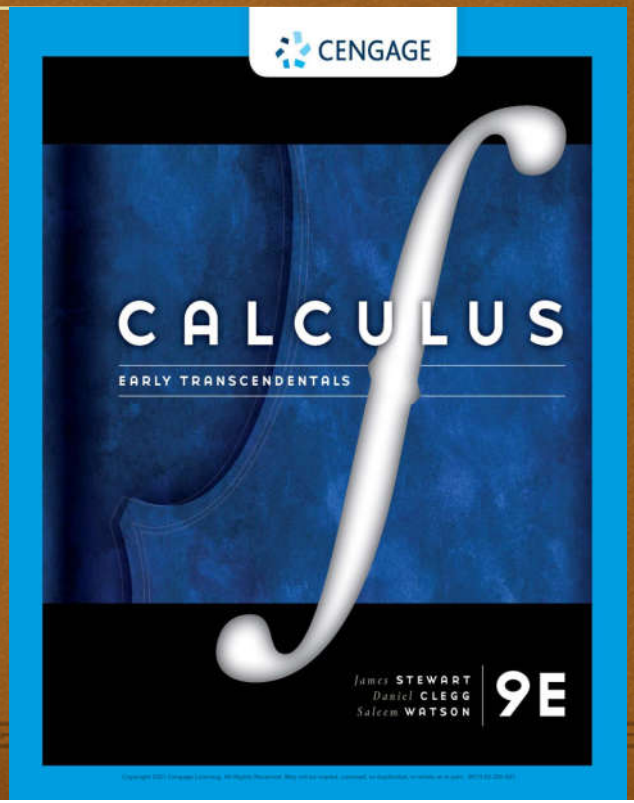


The Integral and Comparison Tests

- In this section, we deal only with series with positive terms, so the partial sums are increasing.

- By the end of this lesson, you will be able to understand about two methods to determine whether a given positive series is convergent or divergent:

The Integral and Comparison Tests.

I. Testing with an integral

Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$

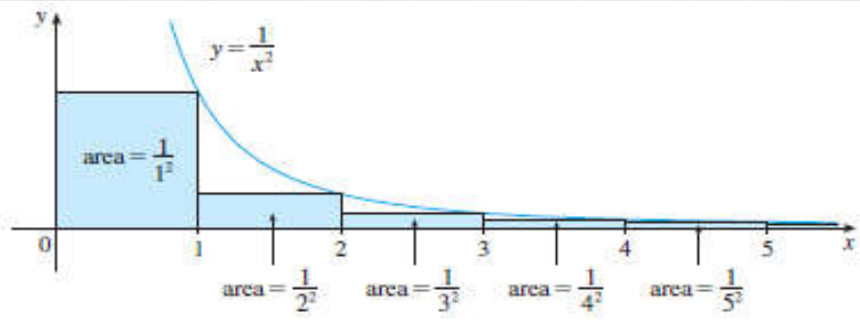
The table of the value of some partial sums is given.

The partial sums are approaching a number near 1.64 as $n \to \infty$. This suggests that the series converges.

n	$s_n = \sum_{i=1}^{n} \frac{1}{i^2}$
5	1.4636
10	1.5498
50	1.6251
100	1.6350
500	1.6429
1000	1.6439
5000	1.6447

Figure 1 shows the curve $y = \frac{1}{x^2}$ and rectangles below the curve.

The sum of the areas of the rectangles is the given series.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} < 1 + \int_1^{\infty} \frac{1}{x^2}\,dx = 2$$

The partial sums are bounded, and the series converges

Testing with an integral

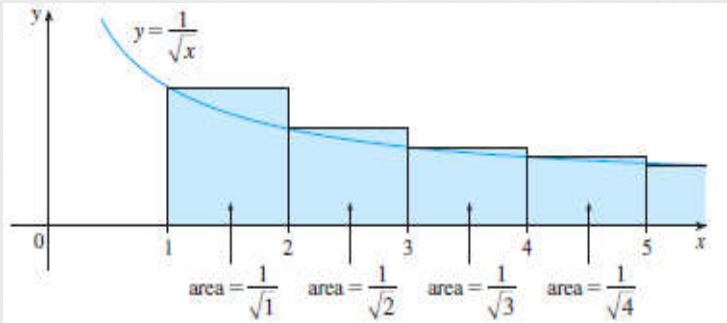
Look at the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots$

The table of values of partial sums suggest that the partial sums are not approaching a finite number, so we suspect that the series may be divergent.

n	$s_n = \sum_{i=1}^{n} \frac{1}{\sqrt{i}}$
5	3.2317
10	5.0210
50	12.7524
100	18.5896
500	43.2834
1000	61.8010
5000	139.9681

Figure 2 shows the curve $y = \frac{1}{\sqrt{x}}$ and rectangles whose tops lie above the curve. Thus

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots > \int_1^{\infty} \frac{1}{\sqrt{x}}\,dx$$



The improper integral diverges, so the series is divergent.

Testing with an integral

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)\,dx$ is convergent. In other words:

(i) If $\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(ii) If $\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

NOTE When we use the Integral Test, it is not necessary to start the series or the integral at $n = 1$. For instance, in testing the series

$$\sum_{n=4}^{\infty} \frac{1}{(n-3)^2} \qquad \text{we use} \qquad \int_4^{\infty} \frac{1}{(x-3)^2}\,dx$$

Also, it is not necessary that f be *always* decreasing. What is important is that f be *ultimately* decreasing, that is, decreasing for x larger than some number N. Then $\sum_{n=N}^{\infty} a_n$ is convergent, so $\sum_{n=1}^{\infty} a_n$ is convergent (see the note at the end of Section 11.2).

Testing with an integral

EXAMPLE 1 Test the series $\sum_{n=1}^{+\infty} \frac{1}{n^2+1}$ for convergence or divergence.

SOLUTION The function $f(x) = 1/(x^2 + 1)$ is continuous, positive, and decreasing on $[1, \infty)$ so we use the Integral Test:

$$\int_1^{\infty} \frac{1}{x^2+1}\,dx = \lim_{t\to\infty} \int_1^{t} \frac{1}{x^2+1}\,dx = \lim_{t\to\infty} \tan^{-1}x\Big]_1^t$$

$$= \lim_{t\to\infty}\left(\tan^{-1}t - \frac{\pi}{4}\right) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Thus $\int_1^{\infty} 1/(x^2 + 1)\,dx$ is a convergent integral and so, by the Integral Test, the series $\sum 1/(n^2 + 1)$ is convergent. ■

Testing with an integral

EXAMPLE 2 For what values of p is the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent?

SOLUTION If $p < 0$, then $\lim_{n\to\infty} (1/n^p) = \infty$. If $p = 0$, then $\lim_{n\to\infty} (1/n^p) = 1$. In either case $\lim_{n\to\infty} (1/n^p) \neq 0$, so the given series diverges by the Test for Divergence (11.2.7).

If $p > 0$, then the function $f(x) = 1/x^p$ is clearly continuous, positive, and decreasing on $[1, \infty)$. We found in Section 7.8 [see (7.8.2)] that

$$\int_1^{\infty} \frac{1}{x^p}\,dx \quad \text{converges if } p > 1 \text{ and diverges if } p \leqslant 1$$

It follows from the Integral Test that the series $\Sigma\, 1/n^p$ converges if $p > 1$ and diverges if $0 < p \leqslant 1$. (For $p = 1$, this series is the harmonic series discussed in Example 11.2.8.) ■

$\boxed{1}$ The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leqslant 1$.

II. Testing by comparing

THE COMPARISON TEST Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

(i) If $\sum b_n$ is convergent and $a_n \leqslant b_n$ for all n, then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ is divergent and $a_n \geqslant b_n$ for all n, then $\sum a_n$ is also divergent.

(i) If we have a series whose terms are ***smaller*** than those of a known ***convergent*** series then our series is convergent.

(ii) If we have a series whose terms are ***larger*** than those of a known ***divergent*** series then our series is divergent.

Testing by comparing

SOLUTION For large n the dominant term in the denominator is $2n^2$, so we compare the given series with the series $\Sigma\, 5/(2n^2)$. Observe that

$$\frac{5}{2n^2 + 4n + 3} < \frac{5}{2n^2}$$

because the left side has a bigger denominator. (In the notation of the Direct Comparison Test, a_n is the left side and b_n is the right side.) We know that

$$\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

is convergent because it's a constant times a p-series with $p = 2 > 1$. Therefore

$$\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$$

is convergent by part (i) of the Direct Comparison Test. ■

Testing by comparing

NOTE Although the condition $a_n \leq b_n$ or $a_n \geq b_n$ in the Direct Comparison Test is given for all n, we need verify only that it holds for $n \geq N$, where N is some fixed integer, because the convergence of a series is not affected by a finite number of terms. This is illustrated in the next example.

EXAMPLE 4

Test the series $\sum_{k=1}^{\infty} \frac{\ln k}{k}$ for convergence or divergence.

Testing by comparing

SOLUTION We used the Integral Test to test this series in Example 11.3.4, but we can also test it by comparing it with the harmonic series. Observe that $\ln k > 1$ for $k \geqslant 3$ and so

$$\frac{\ln k}{k} > \frac{1}{k} \qquad k \geqslant 3$$

We know that $\Sigma\, 1/k$ is divergent (p-series with $p = 1$). Thus the given series is divergent by the Direct Comparison Test. ■

Testing by comparing

THE LIMIT COMPARISON TEST Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

Testing by comparing

EXAMPLE 5

Test the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence or divergence.

EXAMPLE 6

Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}}$ converges or diverges.

BASIC TERMINOLOGIES

1. Integral Test
2. Comparison Test

EXERCISES

- **Exercises: 6 – 32 (P764)**

1.5 – 1.6
Other Convergence Tests

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1.5 Alternating Series and Absolute Convergence

By the end of this lesson, you will be able to understand

1. Alternating series
2. Absolute convergence
3. Conditionally convergent

1.5 Alternating series

An alternating series is a series whose terms are alternately positive and negative.

Example:

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

$$\sum_{n=1}^{\infty}(-1)^{n}\frac{n}{n+1} = -\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \cdots$$

11.5 Alternating series

THE ALTERNATING SERIES TEST If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots \qquad b_n > 0$$

satisfies

(i) $b_{n+1} \leqslant b_n$ for all n

(ii) $\lim_{n \to \infty} b_n = 0$

then the series is convergent.

If the terms of an alternating series decrease to 0, in absolute value, then the series converges

Alternating series

EXAMPLE 1 The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

satisfies

(i) $b_{n+1} < b_n$ because $\dfrac{1}{n+1} < \dfrac{1}{n}$

(ii) $\lim_{n \to \infty} b_n = \lim_{n \to \infty} \dfrac{1}{n} = 0$

so the series is convergent by the Alternating Series Test.

Alternating series

EXAMPLE 2 The series $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ is alternating, but

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{3n}{4n-1} = \lim_{n\to\infty} \frac{3}{4-\frac{1}{n}} = \frac{3}{4}$$

so condition (ii) is not satisfied. Instead, we look at the limit of the nth term of the series:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{(-1)^n 3n}{4n-1}$$

This limit does not exist, so the series diverges by the Test for Divergence.

Alternating series

EXAMPLE 3 Test the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{n^3+1}$ for convergence or divergent?

SOLUTION

(i) For the first condition, we consider the related function

$$f(x) = \frac{x^2}{x^3+1}\ ,\ x \in N.$$

We find that $f'^{(x)} = \frac{x(2-x^3)}{(x^3+1)^2} < 0$ if $x \geq 2$. Thus, f is decreasing when $x \geq 2$. Therefore, $f(n+1) < f(n)$ or $b_{n+1} < b_n$ when $n \geq 2$.

(ii) For the second condition, we find

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n^2}{n^3+1} = 0$$

Thus, the given series is convergent.

Alternating series

ALTERNATING SERIES ESTIMATION THEOREM If $s = \sum (-1)^{n-1} b_n$ is the sum of an alternating series that satisfies

$$\text{(i)}\ 0 \leqslant b_{n+1} \leqslant b_n \qquad \text{and} \qquad \text{(ii)}\ \lim_{n \to \infty} b_n = 0$$

then

$$|R_n| = |s - s_n| \leqslant b_{n+1}$$

The theorem say that the size of the error $R_n = s - s_n$ of the series that satisfies the conditions of the alternating series is smaller than b_{n+1} which is the absolute value of the first neglected term.

Alternating series

EXAMPLE 4 Find the sum of the series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n!}$ correct to three decimal places.

The alternating series satisfies

$$b_{n+1} = \frac{1}{(n+1)!} < \frac{1}{n!} = b_n \qquad \text{and} \qquad \lim_{n\to\infty} b_n = \lim_{n\to\infty}\frac{1}{n!} = 0$$

We have $s = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \cdots$

$$= 1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720} - \frac{1}{5040} + \cdots$$

Notice that $b_7 = \frac{1}{5040} < \frac{1}{5000} = 0.0002$. By the above theorem, we know that $|s - s_6| \le b_7 < 0.0002$. The error of less than 0.0002 does not affect the third decimal place.

Therefore $s \approx s_6 \approx 0.368056$. Thus $s \approx 0.368$.

Absolute convergence

DEFINITION A series $\Sigma\, a_n$ is called **absolutely convergent** if the series of absolute values $\Sigma\, |a_n|$ is convergent.

EXAMPLE 5 The series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n^2}$ is absolutely convergent because $\sum_{n=1}^{\infty}\left|\frac{(-1)^{n-1}}{n^2}\right| = \sum_{n=1}^{\infty}\frac{1}{n^2}$ is a convergent p-series (p = 2).

EXAMPLE 6 The alternating harmonic series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n}$ is convergent but it is not absolutely convergent because $\sum_{n=1}^{\infty}\left|\frac{(-1)^{n-1}}{n}\right| = \sum_{n=1}^{\infty}\frac{1}{n}$ is the harmonic series (p-series, p = 1) and is therefore divergent.

Absolute convergence

DEFINITION A series $\Sigma\, a_n$ is called **conditionally convergent** if it is convergent but not absolutely convergent.

1 THEOREM If a series $\Sigma\, a_n$ is absolutely convergent, then it is convergent.

EXAMPLE 7 The alternating harmonic series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n}$ is conditionally convergent.

EXAMPLE 8 The series $\sum_{n=1}^{\infty}\frac{cosn}{n^2}$ is convergent because the series of absolute value $\sum_{n=1}^{\infty}\left|\frac{cosn}{n^2}\right| \leq \sum_{n=1}^{\infty}\frac{1}{n^2}$ is convergent by comparison test.

11.6 The ratio test

THE RATIO TEST

(i) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).

(ii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(iii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$, the Ratio Test is inconclusive; that is, no conclusion can be drawn about the convergence or divergence of $\sum a_n$.

The ratio test

EXAMPLE 9

Test the series $\sum_{n=1}^{\infty}(-1)^n \frac{n^3}{3^n}$ for absolute convergence.

SOLUTION We use the Ratio Test with $a_n = (-1)^n n^3/3^n$:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\dfrac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{\dfrac{(-1)^n n^3}{3^n}}\right| = \frac{(n+1)^3}{3^{n+1}} \cdot \frac{3^n}{n^3}$$

$$= \frac{1}{3}\left(\frac{n+1}{n}\right)^3 = \frac{1}{3}\left(1+\frac{1}{n}\right)^3 \to \frac{1}{3} < 1$$

Thus, by the Ratio Test, the given series is absolutely convergent. ■

The ratio test

EXAMPLE 10

Test the convergent of series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$

SOLUTION Since the terms $a_n = n^n/n!$ are positive, we don't need the absolute value signs.

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \frac{(n+1)(n+1)^n}{(n+1)\,n!} \cdot \frac{n!}{n^n}$$

$$= \left(\frac{n+1}{n}\right)^n = \left(1 + \frac{1}{n}\right)^n \to e \qquad \text{as } n \to \infty$$

(see Equation 3.6.6). Since $e > 1$, the given series is divergent by the Ratio Test. ■

The Root Test

THE ROOT TEST

(i) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).

(ii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$ or $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(iii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$, the Root Test is inconclusive.

The Root Test

EXAMPLE 11 Test the convergence of the series

$$\sum_{n=1}^{\infty}\left(\frac{2n+3}{3n+2}\right)$$

SOLUTION

$$a_n = \left(\frac{2n+3}{3n+2}\right)^n$$

$$\sqrt[n]{|a_n|} = \frac{2n+3}{3n+2} = \frac{2+\dfrac{3}{n}}{3+\dfrac{2}{n}} \to \frac{2}{3} < 1$$

Thus the given series is absolutely convergent (and therefore convergent) by the Root Test. ■

The Root Test

EXAMPLE 12 Determine whether the following series is convergent or divergent.

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n$$

SOLUTION Here it seems natural to apply the Root Test:

$$\sqrt[n]{|a_n|} = \frac{n}{n+1} \to 1 \qquad \text{as } n \to \infty$$

Since this limit is 1, the Root Test is inconclusive. However, using Equation 3.6.6 we see that

$$a_n = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(\dfrac{n+1}{n}\right)^n} \to \frac{1}{e} \qquad \text{as } n \to \infty$$

Since this limit is different from zero, the series diverges by the Test for Divergence. ■

BASIC TERMINOLOGIES

1. Alternating series
2. Absolute convergence
3. Conditionally convergent

EXERCISES

Exercises: 3 -8, 19 – 40 (P778-779)
41, 42, 43 (P779)